

Visualising Data

120 minutes · 5 self-check questions

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Choosing a chart that shows what the data actually do, and avoiding the plots that hide the very thing a reader needs to see.

Learning objectives

- Match a chart type to the variables being shown
 - Read a histogram, boxplot and scatterplot for shape, spread and outliers
 - Explain why bar charts of means conceal distribution
 - Build publication-quality figures in R and SPSS
 - Apply the conventions that make a figure readable in print and to colour-blind readers
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Which chart for which variables

- One continuous variable: histogram or density plot for shape; boxplot for summary and outliers
 - One categorical variable: bar chart of counts — never a pie chart with more than three slices
 - Continuous by categorical: boxplot, violin plot, or a dot plot showing every observation
 - Two continuous: scatterplot, with a fitted line only if you are claiming a relationship
 - Change over time: line plot with time on the horizontal axis
 - Two categorical: grouped or stacked bar chart, or a mosaic plot
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The bar chart of means problem

- A bar with an error bar shows two numbers and hides the entire distribution
- Radically different datasets — bimodal, skewed, or containing outliers — produce identical bars
- It also implies the bar length is meaningful from zero, which is false for many measurements
- For group comparisons prefer a boxplot, or a dot plot with every point drawn
- With small n , plot every observation; there is no reason to summarise 12 points
- If you must use bars, state whether the error bar is SD, SE or a confidence interval — they differ by a factor of several

IN PRACTICE

Journals in several fields now refuse bar charts of continuous data for exactly this reason. Showing the points costs nothing and lets a reader see what you saw.

Making figures honest

- Start a bar chart axis at zero; truncating it exaggerates differences
- For line and scatter plots a non-zero axis is acceptable, but say so if the range is narrow
- Label both axes with the variable and its unit
- Do not use a second vertical axis to imply a relationship between two series
- Avoid three-dimensional effects entirely — they distort area and length
- Aspect ratio changes the apparent steepness of a slope; keep it consistent when comparing panels

Colour and accessibility

- Roughly 1 in 12 men has some form of colour vision deficiency, most commonly red-green
- Never use red and green as the only distinction between two series
- Use the viridis palette, or vary shape and line type as well as colour
- Check a figure by converting it to greyscale — if it still reads, it will survive printing
- Keep the palette consistent across every figure in a paper or deck
- Use colour to encode data, not for decoration

CHART CHOICE AT A GLANCE

You have	Show shape with	Compare groups with
One continuous variable	Histogram, density plot	Boxplot by group
One categorical variable	Bar chart of counts	Grouped bar chart
Two continuous variables	Scatterplot	Scatterplot coloured by group
Continuous over time	Line plot	One line per group
Small samples ($n < 20$)	Plot every point	Dot plot with median line

Doing it in software

- R base: `hist(dat$bp)`, `boxplot(bp ~ group, data = dat)`, `plot(x, y)`
- ggplot2 gives publication quality: `ggplot(dat, aes(group, bp)) + geom_boxplot() + geom_jitter(width = 0.15)`

- `geom_jitter` over a boxplot shows both the summary and the raw observations
- SPSS: Graphs > Chart Builder, then double-click the chart to edit elements
- Export at 300 dpi or higher, or as a vector format (PDF, EPS, SVG), never a screenshot
- Set the figure size at export rather than resizing afterwards, or the text will not scale with it

Worked examples

Four datasets, one bar chart

Three groups are compared on a continuous outcome. The bar chart of means with error bars looks clean and shows a clear difference. The reviewer asks to see the data.

1. Group A is tightly clustered around its mean.
2. Group B is bimodal — two subgroups, and almost nobody near the mean itself.
3. Group C is skewed with three extreme values pulling the mean up.
4. All three produce a nearly identical bar and error bar.
5. Replacing it with `ggplot(dat, aes(group, y)) + geom_boxplot() + geom_jitter(width = 0.15)` shows all of this at once.
6. The bimodality in group B turns out to be sex, which was the real finding.

THE POINT

A bar chart shows two numbers and hides everything else. Plotting the points costs nothing and sometimes hands you the actual result.

The axis that exaggerated a difference

A slide shows attendance rising from 78% to 82% after an intervention. The bars make it look like a doubling.

1. The vertical axis starts at 76, so the visible bar heights are 2 and 6 — a threefold difference.
2. Bar length encodes magnitude from zero, so the eye reads the ratio of the drawn heights.
3. Redrawn from zero, the bars are nearly the same height, which is the honest picture.
4. If the difference genuinely matters at that scale, a line or dot plot with a restricted axis is acceptable — position, not length, carries the meaning there.
5. Either way, show the confidence intervals: 78% (95% CI 74 to 82) and 82% (95% CI 78 to 86) overlap substantially.
6. The 4-point difference may be real, but the chart was claiming far more than the data support.

THE POINT

Truncating a bar axis is the most common way an honest researcher produces a misleading figure. If the effect only looks impressive on a cropped axis, it is not impressive.

Key terms

Histogram

A chart of the distribution of one continuous variable, using binned counts.

Boxplot

A summary showing median, quartiles and outliers, useful for comparing groups.

Jitter

Small random horizontal displacement so overlapping points become visible.

Truncated axis

An axis not starting at zero, which exaggerates apparent differences in a bar chart.

Viridis

A perceptually uniform colour palette that remains readable in greyscale and to colour-blind viewers.

Vector format

A figure stored as shapes rather than pixels, so it stays sharp at any size.

Activity

Take one finding from your own data and draw it three different ways — for example a bar of means, a boxplot, and a dot plot of every observation. Decide which one a sceptical reader would trust most, and say why the other two are less honest.

Further reading

- **Weissgerber et al. (2015) — "Beyond Bar and Line Graphs"** — PLOS Biology, open access. The paper that changed several journals' policies.
- **Wilke — Fundamentals of Data Visualization** — Free online, and the best single reference on chart choice.
- **ggplot2 documentation and the R Graph Gallery** — Copyable examples for almost any chart you need.

Self-check questions

1. The main criticism of a bar chart showing group means with error bars is that it:
 - a. Is difficult to produce in R
 - b. Hides the distribution, so very different datasets can produce identical bars
 - c. Cannot show more than two groups
 - d. Requires the data to be normally distributed
2. You are comparing a continuous outcome across three groups with 15 participants each. The best chart is:
 - a. A bar chart of means
 - b. A boxplot or dot plot showing every observation
 - c. A pie chart
 - d. A line plot
3. Using only red and green to distinguish two lines is a problem because:
 - a. Those colours print poorly
 - b. Red-green deficiency is the most common form of colour blindness, affecting roughly 1 in 12 men
 - c. Journals charge extra for colour
 - d. It violates copyright
4. A bar chart with a vertical axis starting at 45 instead of 0:

- a. Is standard practice for clarity
 - b. Exaggerates the apparent difference between bars
 - c. Is required when values are large
 - d. Has no effect on interpretation
5. Which R code overlays the raw observations on a boxplot?
- a. `geom_boxplot() + geom_bar()`
 - b. `geom_boxplot() + geom_jitter(width = 0.15)`
 - c. `geom_histogram() + geom_density()`
 - d. `geom_line() + geom_point()`
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Answers

1. **b.** Two numbers cannot describe a distribution. Bimodal data, skewed data and data with an extreme outlier can all yield the same bar and error bar, which is why boxplots or plotting every point are preferred.
 2. **b.** With 15 per group there is no reason to hide the data. A boxplot with jittered points shows the summary and every observation at once.
 3. **b.** A substantial minority of readers cannot separate them. Vary line type or shape as well as colour, and check the figure in greyscale.
 4. **b.** Bar length encodes magnitude from zero. Truncating the axis makes a small difference look dramatic. For line and scatter plots a non-zero axis is acceptable, because position rather than length carries the meaning.
 5. **b.** `geom_jitter` displaces points slightly so overlapping observations become visible. Layered over a boxplot it gives both the summary and the underlying data.
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